NARAYANA IIT ACADEMY INDIA

46 YEARS OF EXCELLENCE

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 30-04-26
Time: 3 Hrs Max. Marks: 180

30-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-10(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 7 – 12)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 13 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 25)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 26 – 31)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 32 – 35)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 44)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 45 – 50)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

1. The equation $log_x 216 + log_{2x} 64 = 3$ has

A) one irrational solution
B) no solution as a prime number
C) two real solutions
D) one integral solution

2. Let $f(x) = 2\sin^2\beta + 4\cos(x+\beta)\sin x\sin\beta + \cos 2(x+\beta)$ and

$g(x) = \left(\lim_{\alpha\to 0}(1+\alpha)^{\frac{1}{\alpha}}\right)^{\tan\left(\tan^{-1}x\right)}$ Then

A) number of integers in the range of $f(x)\forall x \in R$, is 3

B) $\lim_{x\to 0}(f(x))^{\frac{1}{x^2}}$ is equal to e^{-2}

C) $\lim_{x\to 0}\frac{f(x)-g(x)}{x}$ is equal to -1

D) number of solution of the equation $g(x) = 1 - x$ is 2

3. If $y = \sqrt[3]{6xy^2 - 11x^2y + 6x^3}$, then possible value(s) of $\frac{dy}{dx}$ at $x = 2$ is(are)

A) 0 B) 1 C) 2 D) 3

4. The common tangent(s) of the circle $(x+1)^2 + y^2 = 1$ and the parabola $y^2 = 4x$ is (are)

A) $x = 0$ B) $y = \frac{1}{\sqrt{3}}x + \sqrt{3}$ C) $y = \frac{-1}{\sqrt{3}}x - \sqrt{3}$ D) $y = x$

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5. Let P be a point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. The tangent drawn at P meets the coordinate axes at A and B , then(where O is origin)

A) minimum area of ΔOAB is 6 B) minimum area of ΔOAB is 12

C) minimum length of AB is 5 D) minimum length of AB is $\sqrt{13}$

6. The eccentricity of hyperbola satisfying the differential equation $2xy\frac{dy}{dx}=x^2+y^2$, and passing through $P(2,1)$, is

A) $2\sqrt{2}$ B) $\sqrt{2}$ C) $3\sqrt{2}$ D) 2

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 7 and 8:

Let A and B are inverse matrices of each other of order 3 such that $A(B+A)=2I$.

7. The value of det. $\left((2\ \text{adj.A})^2+3(\ \text{adj.B})^2\right)$ is equal to (where $\text{adj}(M)$ denotes adjoint and $\det(M)$ denotes determinant value of matrix M respectively)

8. If $P=\left(A^{-1}+B^{-1}\right)$ (adj. $(2A)$+ adj. $(2B)$) then absolute value of trace of P , is (where $\text{adj}(M)$ denotes adjoint and M^{-1} denotes inverse of matrix M respectively)

Question Stem for Question Nos. 9 and 10:

Let $f(x)=x^3-6\sin\theta\cdot x^2+\left(2\sin^2\theta+q\right)x-1; q,\theta\in R$ be a cubic polynomial in x. If exactly one tangent can be drawn to the graph of $y=f(x)$ which is parallel to the line $y=x$, then

9. The largest value of q is

10. If q is smallest, then the value of $\frac{d}{dx}\left(f^{-1}(x)\right)\Big|_{x=9}$ is

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Question Stem for Question Nos. 11 and 12:

Let A and B be two events such that $P(A)=\frac{1}{4}, P(B)=\frac{1}{2}$ and $P(A \cup B)=\frac{3}{5}$

11. $P(A \cap B)$ is

12. $P(A/B)$ is :

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 13 and 14:

Let $x, \frac{[x]}{2}$ and $\{x\}, (x>0)$ are the first three terms of a G.P.

(where $[k]$ and $\{k\}$ denote greatest integer and fractional part functions of k respectively)

13. Common ratio of the G.P. is

A) $\tan\left(\frac{\pi}{12}\right)$ B) $\tan\left(\frac{\pi}{6}\right)$ C) $\tan\left(\frac{\pi}{8}\right)$ D) $\tan\left(\frac{3\pi}{8}\right)$

14. If $1<x<2$ then product of first three term of the G.P. is

A) $\frac{1}{8}$ B) $(\sqrt{2}-1)^3$ C) $\frac{1}{4}(\sqrt{2}-1)^2$ D) $\frac{1}{4}$

Question Stem for Question Nos. 15 and 16:

Consider, three vectors

$\vec{v}_1 = sin\theta\hat{i} + cos\theta\hat{j} + (a-3)\hat{k}$

$\vec{v}_2 = (sin\theta + cos\theta)\hat{i} + (cos\theta - sin\theta)\hat{j} + (b-4)\hat{k}$

$\vec{v}_3 = cos\theta\hat{i} - sin\theta\hat{j} + (c-5)\hat{k}$

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15. If $\vec{v}_1+\vec{v}_2+\vec{v}_3=\lambda\hat{i}$ where $\theta\in[0,2\pi]$ and $a,b,c\in N$ then number of quadruplets (a,b,c,θ) are

A) 55 B) 91 C) 110 D) 182

16. If $\vec{v}_1,\vec{v}_2,\vec{v}_3$ represent sides of a triangle PQR in xy -plane and corresponding to triangle PQR there exist one more triangle ABC whose sides are denoted by a,b,c, then $\frac{\text{Ar}(\Delta ABC)}{\text{Ar}((\Delta PQR))}$ is equal to:

A) 6 B) 9 C) 12 D) 24

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

17. If $\sum_{n=0}^{\infty}tan^{-1}\left(\frac{cot^{-1}(n^2+3n+3)}{1+cot^{-1}(n+1)cot^{-1}(n+2)}\right)'=tan^{-1}\left(\frac{p}{4}\right)$ then the value of $\left[cos^{-1}(cos(p-1))\right]$ is____ (where $[k]$ denotes greatest integer less than or equal to k)

18. Let $f(x)=\sqrt{6-5^{\sin^{-1}x}-5^{1-\sin^{-1}x}}$ and $g(x)=\sqrt{5^{\sin^{-1}x}}$ and if the complete set of range of the function $h(x)=f(x)\cdot g(x)$ is $[a,b]$, then $(a+b)$ equals

19. The value of $\sum_{r=0}^{19}r(r+1)\frac{^{20}C_{r+1}}{^{20}C_r}$, is

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PHYSICS **MAX.MARKS: 60**

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Negative Marks: -2 In all other cases.

20. A high-resolution oil-immersion microscope is used to observe two closely spaced protein filaments in a biological sample.

Given

- Wavelength used in air : $\lambda_0 = 480\ \text{nm}$
- Refractive index of immersion oil : $\mu = 1.60$
- Semi-angle of acceptance of objective : $\alpha = 74^\circ$

Later, the same microscope is used with blue light of wavelength $\lambda_0 = 420\ \text{nm}$ keeping everything else unchanged.

Which of the following statements are correct?

A) The minimum resolvable distance using $\lambda_0 = 480\ \text{nm}$ is about $0.16\mu\ m$

B) Two filaments separated by $0.18\mu\ m$ can be resolved with 480 nm light but not with 420 nm light.

C) Switching from 480 nm to 420 nm improves the resolving power by more than 14% .

D) If the immersion oil is replaced by air ($\mu = 1$), the minimum resolvable distance becomes larger than 0.50μ m

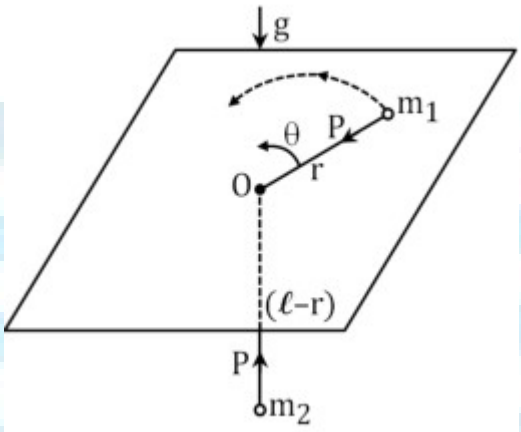
21. A YDSE experiment is used to measure the wavelength of light used. The distance between screen and slits is measured by a meter scale with least count 1 mm . The measured distance is 1.500 m . The distance between the slits is measured by a micrometer of least count 0.01 mm . The distance between slits is measured to be 0.33 mm . The fringe width is measured by the same micrometer. It comes out to be 2.50 mm . The wavelength of light is :-

A) 550 ± 19 nm B) 550 ± 39 nm C) 330 ± 28 nm D) 330 ± 56 nm

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22. A particle of mass m_1 moves on a frictionless, horizontal plane. It is connected to a second particle of mass m_2 by an inextensible string which passes through a small hole at point O, as shown in figure. If the second mass moves only along a vertical line through O, and if the initial conditions are $r(0)=r_0, \dot{r}(0)=0, \dot{\theta}(0)=\omega_0$.

(Take : $m_1=m_2=m$ and $r_0\omega_0^2=\frac{g}{3}$)



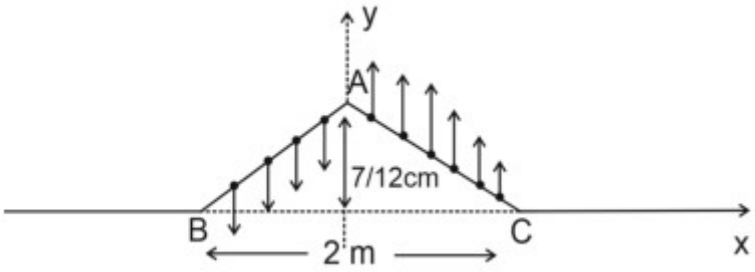
A) The minimum value of r is $\frac{r_0}{2}$.

B) The minimum value of r is $\frac{r_0}{3}$.

C) The maximum value of tension in the string is $\frac{7}{6}mg$.

D) The maximum value of tension in the string is $\frac{11}{6}mg$.

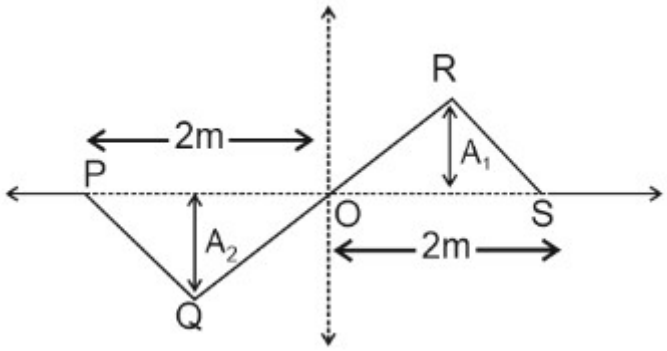
23. Figure shows the shape of string at an instant $t=0$. Tension in the string is $T=12$ N and mass per unit length of the string is $\mu=1/12$ kg / m .



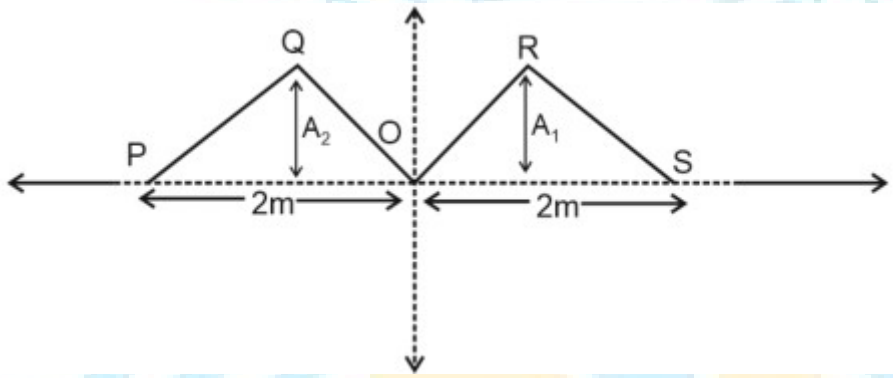
Velocity of all the particles lying on right half section AC of string is 25 cm / s upwards and velocity of all the particles lying on left half section AB of string is 25 cm / s downwards. Assume that triangular wave pulses can smoothly propagate along the string. Case 1 and Case 2 represents two possible cases of what the shape of the string will look like at $t=1/12$sec.

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Case-1 :



Case-2 :



Which of the following statement(s) are correct?

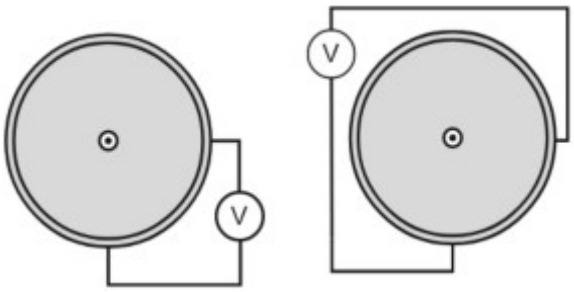
A) Case-2 represents the correct shape of string and $A_1 = \frac{5}{12}$ cm and $A_2 = \frac{1}{6}$ cm

B) Case-1 represents the correct shape of string and $A_1 = \frac{4}{3}$ cm and $A_2 = \frac{3}{4}$ cm

C) Case-1 represents the correct shape of string and speed of particles in section PQ is 9 cm / sec and speed of particles in section RS is 16 cm / s

D) Case-2 represents the correct shape of string and speed of particles in section PQ is 16 cm / sec and speed of particles in section RS is 9 cm / s

24. In a cylindrical region that is encircled by a metallic ring of uniform cross sections, uniform magnetic field perpendicular to the plane of ring varies with time in such a manner that a constant electromotive force of 12 V is induced in the ring. A voltmeter of very high resistance is used to measure potential difference between two points on the ring. For the purposes, the voltmeter is connected in two different manners as shown in figures. In both arrangements, the ring and connecting wires are in same plane. Resistance of left half of ring is twice that of right half of the ring.

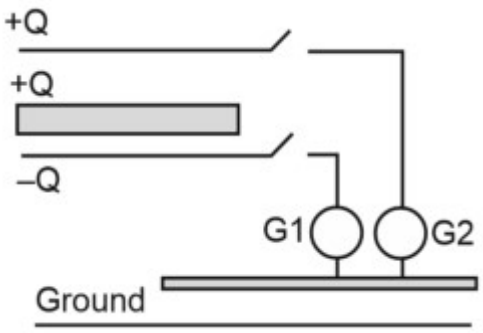


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Which of the following conclusion can you draw regarding V_I and V_{II} of the voltmeter in the first and second arrangement respectively?

A) $V_I = 3\ V$ B) $V_{II} = 9\ V$ C) $V_I = 2\ V$ D) $V_{II} = 10\ V$

25. A parallel-plate capacitor is charged as shown (Q is given). A metal slab with the total charge +Q is placed inside the capacitor as shown. The thickness of the slab is d . The distance between the top plate and the top of the slab is 2 d , and the distance between the bottom plate and the bottom of the slab is d . Each plate is grounded through a galvanometer as shown. Find the charge that passes through each galvanometer after both switches are closed simultaneously.



A) A total charge of $+Q/3$ from through G1 from the ground to bottom plate.

B) A total charge of $Q/3$ from through G1 from the bottom plate to the ground.

C) A total charge of $4Q/3$ from through G2 from the top plate to the ground.

D) A total charge of $4Q/3$ from through G2 from the ground to top plate

SECTION - 2 (Maximum Marks : 12)

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Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 26 and 27:

A heat pump working on the Carnot cycle takes in heat from a reservoir at $5^\circ C$ and delivers heat to a reservoir at $60^\circ C$. The heat pump is driven by a reversible heat engine which takes in heat from a reservoir at $840^\circ C$ and rejects heat to a reservoir at $60^\circ C$. The reversible heat engine also drives a machine that absorbs 30 kW . The heat pump extracts 17 kJ / s from the $5^\circ C$ reservoir.

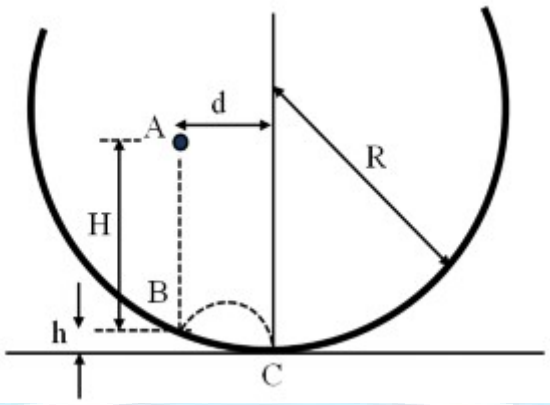
26. The rate of heat supply from the $840^\circ C$ source (in kW) is

27. The rate of heat rejection to the $60^\circ C$ sink(in kW) is :-

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Question Stem for Question Nos. 29 and 29:

A small marble is released vertically with zero initial velocity onto a spherical surface of radius R . The marble is dropped from a height H (point A in the figure), measured from the spherical surface, and at a distance d from the axis of symmetry of the spherical surface, as shown in the figure. The marble rebounds elastically from the surface (point B) and falls again exactly at the center of the surface (point C).



28. The time taken by the marble, to rebound from the surface (point B) until it falls at the center of the surface (point C), is $\frac{1}{\alpha}\frac{R^2}{\sqrt{gH}(R-h)}$. The value α is

29. Consider now that the marble is released from a distance very, very close to the axis of symmetry. In this case, calculate the approximate value of the height H , in terms of the radius R of the spherical surface, from which the marble must be released. If value of H is $\frac{R}{\beta}$, the value of β is

Question Stem for Question Nos. 30 and 31:

Two coherent light sources simultaneously emit two monochromatic radiations with wavelengths $\lambda_1 = 600$ nm and $\lambda_2 = 0.4\mu$ m. The distance between the sources is $s = 1$ mm, and on a screen located at a distance $d = 1.2$ m from the plane of the sources, parallel to it, an interference pattern is observed. At the center of the screen, a square of side $\ell = 6$ mm is marked such that the central maximum lies at the center of the square, and the fringes are parallel to two of its sides.

30. The minimum distance from central maxima on the surface of the square where the bright fringes coincide, (in mm) is :-

31. A third light source, identical to the first two, is placed midway between them. In front of the three sources, an absorption filter for radiation of wavelength λ_2 is placed. The intensity ratio of the two types of bright fringes observed is

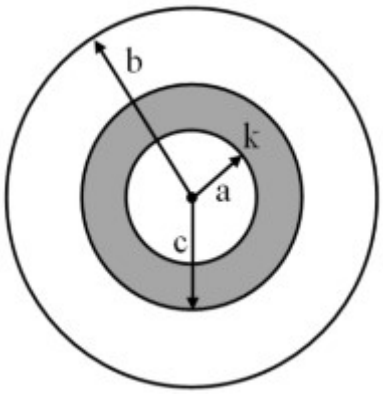
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SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
For each question, choose the option corresponding to the correct answer.
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Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 32 and 33:

A capacitor is made of two concentric cylinders of radius a and $b\,(a < b)$, and length $\ell \gg b$. The region between the cylinders is filled with a cylindrical layer of dielectric k, inner radius ' a ' and outer radius $c = \sqrt{ab}$ as shown in the figure.



32. Find the capacitance of the system.

A) $\dfrac{4\pi k\varepsilon_0\ell}{(1+k)\ell n\left|\dfrac{b}{a}\right|}$

B) $\dfrac{4\pi k\varepsilon_0\ell}{(1+k)\ell n\left|\dfrac{a}{b}\right|}$

C) $\dfrac{2\pi k\varepsilon_0\ell}{(1+k)\ell n\left|\dfrac{b}{a}\right|}$

D) $\dfrac{4\pi k\varepsilon_0\ell}{(k)\ell n\left|\dfrac{b}{a}\right|}$

33. Find the work done to remove the dielectric cylinder while maintaining the potential difference v between the two cylinders constant.

A) $\dfrac{(k+1)}{(k-1)}\dfrac{\pi\varepsilon_0\ell\ V^2}{\ell n\left|\dfrac{b}{a}\right|}$

B) $\dfrac{(k-1)}{(k+1)}\dfrac{\pi\varepsilon_0\ell^2\ V}{\ell n\left|\dfrac{b}{a}\right|}$

C) $\dfrac{(k-1)}{(k+1)}\dfrac{\pi\varepsilon_0\ell\ V^2}{\ell n\left|\dfrac{b}{a}\right|}$

D) $\dfrac{(k-1)}{(k+1)}\dfrac{\pi\varepsilon_0\ell^2\ V^2}{\ell n\left|\dfrac{b}{a}\right|}$

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Question Stem for Question Nos. 34 and 35:

A pendulum is suspended from a fixed point. The mass of the bob of the pendulum is m and the length of string is ℓ. It is initially at rest at its lowest position. A particle of mass $\frac{m}{2}$ moving horizontally along negative x -direction with velocity $\sqrt{2\,g\ell}$ collides with the bob and comes to rest. When the bob comes to rest, another particle Q of mass m moving horizontally along y -direction (perpendicular to initial direction of mass $\frac{m}{2}$) collides with the bob and sticks to it. It is observed that the bob now moves along a horizontal circle. There is horizontal floor 2ℓ distance below the point of suspension of the pendulum.

34. Which of the following is possible value of speed of particle Q just before its collision with the bob?

A) $\sqrt{\frac{7\ell\,g}{3}}$ B) $\sqrt{\frac{3\ell\,g}{4}}$ C) $\sqrt{\frac{7\ell\,g}{4}}$ D) $\sqrt{\frac{4\ell\,g}{3}}$

35. If string breaks during the circular motion, the bob will hit the floor after a time equal to:

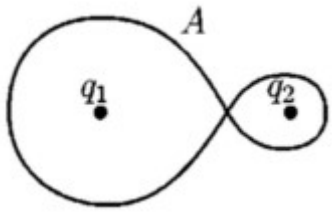
A) $\sqrt{\frac{4\ell}{g}}$ B) $\sqrt{\frac{4\ell}{2\,g}}$ C) $\sqrt{\frac{5\ell}{2\,g}}$ D) $\sqrt{\frac{\ell}{2g}}$

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SECTION - 4 (Maximum Marks : 12)

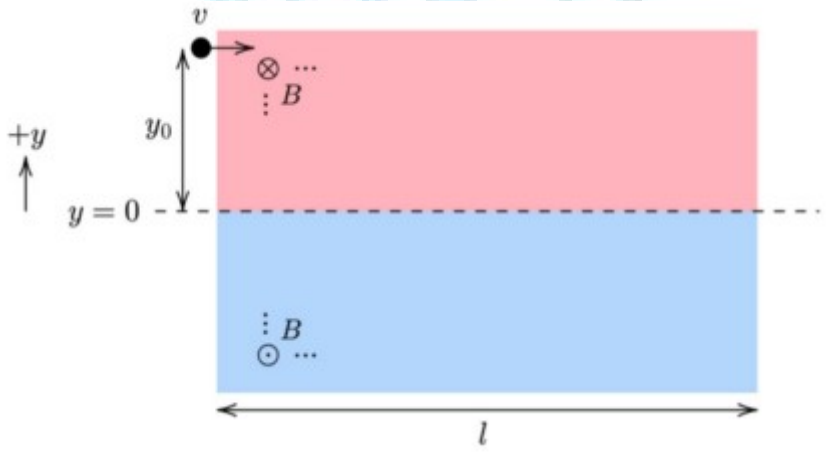
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
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Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

36. At all points of curve A shown in the figure, the potential of the electric field created by two stationary points charges $q_1 = 4nC$ and $q_2 = 1nC$ is equal to $\varphi = 900\ V$. Determine the distance ℓ (in cm) between the charges. The constant in Coulomb's law is $k = 9\times10^9\,Nm^2/C^2$.

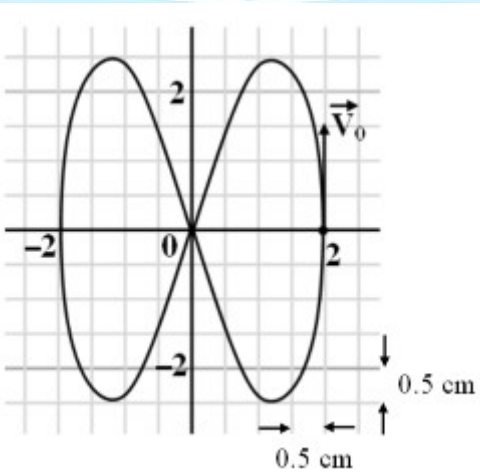


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37. Consider a magnetic field which extends infinitely along the vertical y -axis, but has a finite length $\ell = 870$ mm along the horizontal x -axis. The magnetic field has uniform magnitude B and points into the page for $y > 0$, but out of the page for $y < 0$. An electron is initially located at y -coordinate $y_0 = +170$ nm. It is propelled horizontally into the magnetic field with a velocity $v = \frac{y_0 eB}{m_e}$, where e is the elementary charge, and m_e is the mass of an electron. Find the y -coordinate (in mm) of the electron, y_1, when it exits the field. Neglect gravity. The value of $|y|$ is



38. A body, which can be considered a point, participates in two elastic interactions along the directions of the axes Ox and Oy both lying in the horizontal plane. Due to these interactions, the body describes the planar trajectory shown in the adjacent figure. The values on the axes are expressed in centimeters. At the moment $t = 0$, the velocity of the body is directed along the Oy axis and has the value $v_0 = 10$ cm/s. The angular frequency of oscillation (in rad/sec) along the y -axis is :



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CHEMISTRY **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

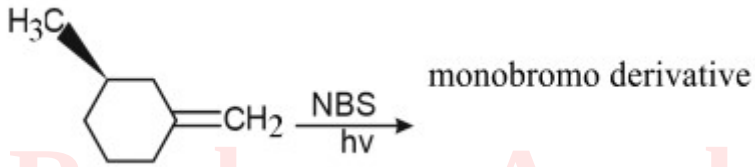
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. Which of the following reaction is not an example of redox reaction?

A) Hydrolysis of XeF_2 B) Hydrolysis of XeF_6

C) Reaction of XeF_4 with O_2 F_2 D) Reaction of XeF_2 with PF_5

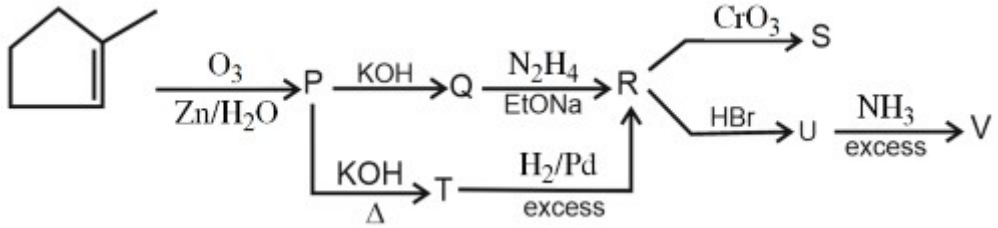
40. Observe the following reaction:



Pick the correct statement (s)

A) 6 optically active products are formed
B) All products formed in the reaction show geometrical isomerism
C) 4 chiral and 2 achiral products are formed
D) All products formed are optically active

41.



Which of the following statement(s) is(are) correct based on above sequence of reaction?

A) V is less basic than aniline but it gives carbyl amine reaction
B) V is more basic than pyridine but doesn't give Mustard oil reaction
C) S will not give coloured ppt with 2,4-DNP but gives Fehling's test
D) S on reduction with H_2/Ni gives a product which gives Lucas test

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42. $XO_4^- + X^{2+} + OH^- \rightarrow XO_2 + H_2O$ (unbalanced)

$XO_2 + HY \rightarrow XY_2 + H_2O + Y_2$ (unbalanced)

Information-1: Y_2 is a greenish yellow gas, due to π^* – HOMO $\rightarrow$ σ^* - LUMO transition.

Information-2: X^{2+} (aq) gives an acid on reaction with $NaBiO_3 + HNO_3$. This acid is coloured due to LMCT.

Based on above information which of the following options are correct

A) When XO_4^- is treated with conc. H_2SO_4 then green colour compound is formed.

B) On passing limited Y_2 to aqueous KI results violet solution in $CHCl_3$

C) 1 mole of XO_4^- is enough to oxidise 5 mole of $NaHC_2O_4$

D) Y_2 (excess) $+Na_2S_2O_3 + H_2O \rightarrow HY + NaHSO_4$ (unbalanced)

43. Which among the following statement(s) is (are) true for the extraction of aluminium from bauxite?

A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.

B) Addition of Na_3AlF_6 lowers the melting point of the electrolyte mixture.

C) CO_2 is evolved at the anode during electrolysis.

D) The cathode is a steel vessel with a lining of carbon.

44. Identify the option(s) of CORRECT statement(s) regarding synergic bonding in molecules-

(I) The C-O bond order among iso electronic and iso-structural species $[Ni(CO)_4] > [Co(CO)_4]^- > [Fe(CO)_4]^{2-}$

(II) Metal carbonyls have large tendency to follow Sidgwick E.A.N. rule

(III) The metal-carbon bond order in $[Me_3PNi(CO)_3]$ is less than in $[F_3PNi(CO)_3]$

(IV) The paramagnetic character of the molecule/ion remains constant during synergic bonding.

(V) If CO is replaced by NO in $[Cr(CO)_6]$, then it is better represented as $[Cr(NO)_4]$, rather than $[Cr(NO)_6]$

A) I, II B) I, III C) III, IV D) IV, V

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SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

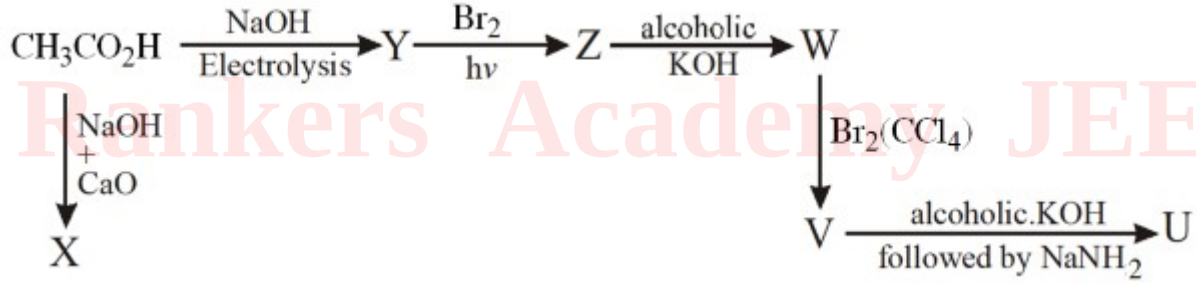
Question Stem for Question Nos. 45 and 46:

A volume of 100 ml of $0.1M-H_3PO_4$ solution is titrated with $0.5M-NaOH$ solution till the second equivalence point. Then 10 ml of $0.5MHC\ell$ solution is mixed in the resulting solution. The dissociation constants of H_3PO_4 are $10^{-3},10^{-8}$ and 10^{-13}.

45. pH at the second equivalence point is ____

46. pH of the solution after adding $HC\ell$ is ____

Question Stem for Question Nos. 47 and 48:



47. $U \xrightarrow[H_2SO_4]{1\%HgSO_4} (A) \xrightarrow{SeO_2} (B) \xrightarrow[(ii)\ H^+]{(i)\ conc.\ KOH} (C) \xrightarrow{\Delta} D$

DOU in D is

48. Sum of DOU in X, W & U is

Question Stem for Question Nos. 49 and 50:

Primary structure of protein is the linear poly peptide chain of amino acids, it contain peptide linkage between amino acids.

The sequence of amino acids is specific for a particular protein such as insulin hormone is also a protein which contains 51 amino acids. Complete acidic hydrolysis of insulin require $50H_2O$ molecules which hydrolyse 50 peptide bonds.

49. How many H_2O molecules are required for complete hydrolysis of decapeptide chain.

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50. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. Sum of total number of units of alanine and phenylalanine present in the decapeptide is.

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 51 and 52:

For the first order dissociation reaction :

$A_2B_3(aq.) \rightarrow 2\,A^{3+}(aq.) + 3B^{2-}(aq.)$

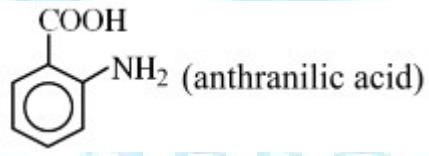
a definite volume of solution is taken at different instants and its osmotic pressure rise is measured. If osmotic pressure at $[t=0]$ is 2 mm of Hg and at $[t=10]$min is 6 mm of Hg then answer the questions given below for the solution:
[Given density of solution remains constant and is equal to $1gm/ml$]

51. Osmotic pressure of solution in mm of Hg at $t = 20$ min is:
A) 7 mm B) 8 mm C) 10 mm D) 12 mm

52. Time required for 87.5% of the reaction to complete is:

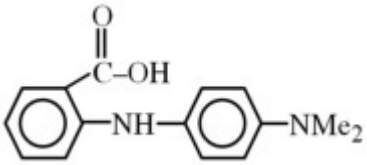
Question Stem for Question Nos. 53 and 54:

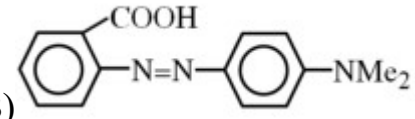
Methyl red is commonly used as indicator for acid base titrations. It is prepared by treating $NaNO_2/HCl$ with anthranilic acid and the resulting solution is mixed with N,N dimethyl aniline and shaked well the solution for some minutes to get "Methyl Red".

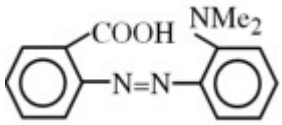
Given : 

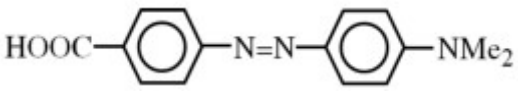
(anthranilic acid)

53. Which is most likely to be "Methyl Red"?

A) 

B) 

C) 

D) 

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54. Which is the correct option regarding anthranilic acid?

A) It is a natural amino acid
B) With sodalime it gives aniline
C) It gives -ve carbyl amine test
D) It is not aromatic in nature

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

55. Total number of basic radicals which give yellow ppt. with $(NH_4)_2\ S_{(aq)}$.

$Pb^{+2}, Ag^{+}, Cd^{+2}, As^{+3}, As^{+5}, Zn^{+2}, Mn^{+2}, Ni^{+2}, Co^{+2}$

56. A sample of excited single electron ions can emit maximum 10 different spectral lines when de-excited to ground level. The maximum energy emitted during transition is 117.5 eV . Now, photons of minimum energy emitted during transition strikes on a metal having work function 1.254 eV emits photoelectrons. The de-Broglie wavelength of emitted photo electron (in Å) is. (Answer to the nearest integer)

57. A sample of water is electrolysed during an experiment using 40 F electricity with 75% current efficiency. The released O_2 is used in a bomb calorimeter to burn $3.5\ g C_2H_4$ gas at 300 K . The temperature of the calorimeter was found to increase from 300 K to 300.50 K due to combustion process. The magnitude of heat of combustion at constant pressure

(in kJ) for 5 moles of ethene is:

(Given: Heat capacity of the calorimeter system is $3\ kJ/K, R = 8\ J/mol.K$)

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